main.py - Plain-English Line-by-Line Guide

What this file does (big picture)

main.py is the front door of your app. When you run it, Python starts at the top and walks downward, one statement at a time.

In plain English, today's main.py does these jobs:

1) Import helpers (config, ai_client, guardrails, knowledge, orders, responses, support)
2) Show a title, confirm the API key, and show AI client provider/model
3) Run input guardrails BEFORE classification
4) Classify into a support lane
5) Collect facts (orders / knowledge), then responses.py writes customer language
6) Run output guardrails on the draft, then print
7) Escalate with an EscalationPacket when needed

Professional habit: facts vs wording are separate. orders/knowledge discover truth; responses.py communicates it (template + AI warm rewrite); guardrails enforce safety.

New words you will see in this file

import	Borrow code that someone (including you) wrote in another file.
guardrail	A hard safety check that can allow, block, or escalate a message.
facts vs wording	What is true (data) vs how we say it (responses.py + AI).
warm rewrite	AI rephrases factual drafts into friendlier language.
SystemExit	Stop the program early on purpose (after a block or escalate).

Flow while the program runs

Ask question -> check_input -> classify ->
get facts -> responses.render_* (AI warm rewrite) ->
apply_output_guardrails -> print.

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1

```
from ai_client import describe_ai_client
```

Plain English:
Borrow a safe one-line AI connection status (provider + model, never the key).

2

```
from config import print_openai_key_status
```

Plain English:
Borrow the safe API-key checker from config.py.

3-7

```
from guardrails import ...
```

Plain English:
Borrow input/output safety tools.

8-9

```
from knowledge import search_knowledge / from orders import get_order_help_facts
```

Plain English:
Borrow *fact* finders only - not customer prose.

10-16

```
from responses import render_*
```

Plain English:
Borrow the writer that turns facts into customer-facing language.

17-27

```
from support import ( ... )
```

Plain English:
Borrow categories, classifier, and escalation packet builder.

29-39

```
banner + key_ok + describe_ai_client()
```

Plain English:
Title block, prove the secret loaded, then show which AI provider/model is configured.

39-41

```
question = input(...) ; print what they asked
```

Plain English:
Collect the customer message and echo it back.

43-55

```
safety = check_input(...) ; escalate or block exits
```

Plain English:
Safety first. Non-allow actions print a safe message and stop.

57-60

```
classify_request + category_label + print
```

Plain English:
Decide the support lane only after guardrails allow.

64-80

```
facts then render_*(..., question=question)
```

Plain English:
Pass the customer question into responses so the AI warm rewrite has context. Sleep disclaimer stays forced for sleep lane.

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81-86

```
ESCALATION: render_escalation_customer + packet
```

Plain English:
Customer handoff language (possibly AI-warmed); operator packet from support.py.

88

```
print(apply_output_guardrails(draft, question=question))
```

Plain English:
Final safety net on AI-warmed wording (refund promises, leaks, etc.).